On-Device Speech Recognition on iOS

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SpeechAnalyzer (iOS 26+)

SpeechAnalyzer is Apple's newer, actor-based, fully on-device speech-to-text framework introduced in iOS 26. It has no one-minute cap and uses a modern async API. [11] [9]

It requires language-specific assets to be present on the device. Availability should be checked, with DictationTranscriber as a fallback, which supports the same languages, model and devices as the on-device SFSpeechRecognizer. [3] [2]

SFSpeechRecognizer (legacy, iOS 10+)

If supportsOnDeviceRecognition is false, the recognizer needs a network. Setting requiresOnDeviceRecognition to true keeps audio off the network, but it is only honored when on-device support is available. [7] [6]

Without requiresOnDeviceRecognition set to true, Apple may use its online recognition service. [10]

Practical considerations

Supported does not mean installed: a third-party write-up notes the on-device model for a language arrives only when the system fetches it. [5]

Apple's live-audio sample code does not run in the iOS Simulator and requires a physical device running iOS or iPadOS 27 or later. [8]

One comparison recommends SpeechAnalyzer on iOS 26+ and WhisperKit on earlier versions. One developer reported 0.86 s for a transcription on an A14 iPhone; this is a single anecdotal test. [12] [4]

A 2026 blog post says long-form transcription with custom vocabulary still needs SFSpeechRecognizer until SpeechTranscriber supports contextual strings. This is unverified against Apple's docs. [1]

Sources

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